CS 412 / 512
Introduction to Computer Graphics

Zhu Wang
Assistant Professor of Computer Science



COMPUTER SCIENCE

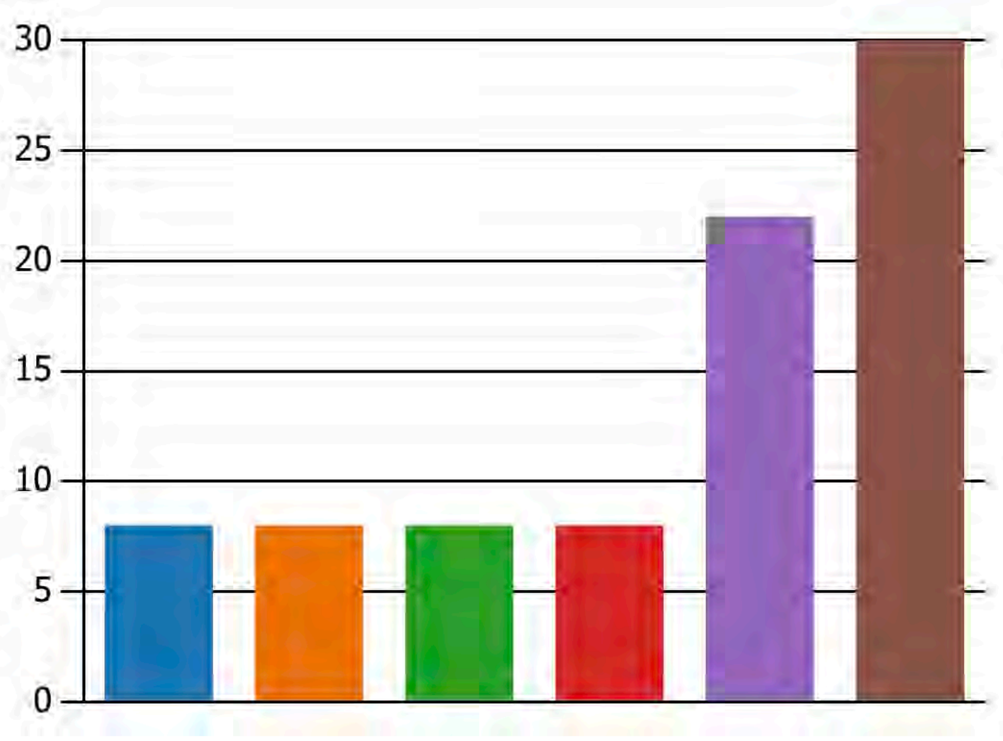
Quiz 2 Review

Affine Transformation

Affiine preserves straight lines and parallelism

Which one/ones of the following are not affine transformation?

Scaling transformation	8
Rotation transformation	8
Translation transformation	8
Reflection/Mirroring transforma...	8
Orthographic transformation	22
Perspective transformation	30

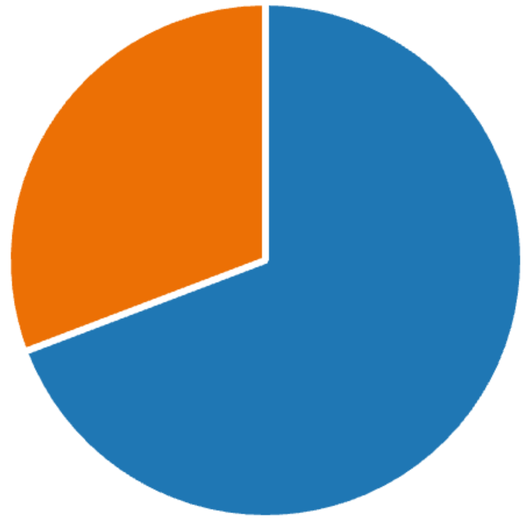


Homogeneous Coordinates

It is a vector when $w = 0$

A translation matrix moves objects by (3, 0, 0). It is applied to two homogeneous vectors: **a**= (1, 2, 3, 1) **b** = (1, 2, 3, 0), what are the results **a′** and **b′**?

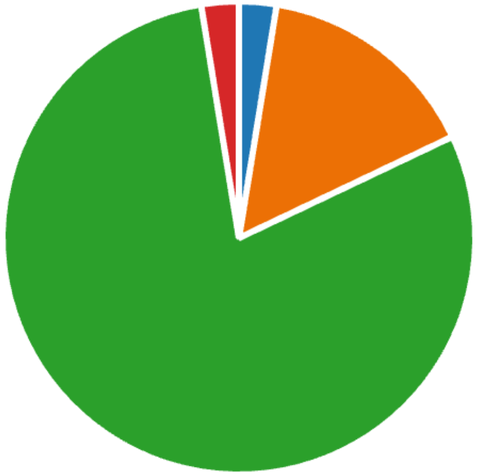
- **a'**=(4, 2, 3, 1), **b'**=(4, 2, 3, 0) — 27
- ✓ **a'**=(4, 2, 3, 1), **b'**=(1, 2, 3, 0) — 12
- **a'**=(1, 2, 3, 1), **b'**=(4, 2, 3, 0) — 0
- **a'**=(4, 2, 3, 1), **b'**=(3, 0, 0, 0) — 0



Lighting & Shading

In the Phong reflection model, which one depends on the viewer's position?

	Ambient	1
	Diffuse	6
✓	Specular	31
	None of the above	1

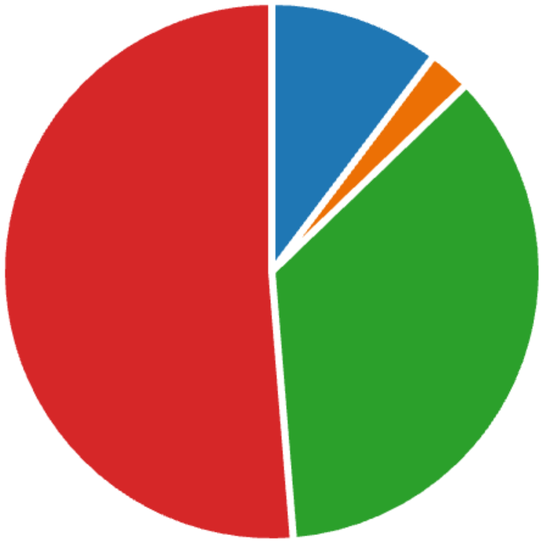


$$I = I_a K_a + I_d K_d (l \cdot n) + I_s K_s (v \cdot r)^{\alpha}$$

Lighting & Shading

Which controls the shininess/sharpness of a surface reflection in Phong shading?

Ambient reflection coefficient k_a	4
Diffuse coefficient k_d	1
Specular reflection coefficient k_s	14
Specular exponent n, which raises (R · V) to a power	20



$$I = I_a K_a + I_d K_d (l \cdot n) + I_s K_s (v \cdot r)^{\alpha}$$

Lighting & Shading

In the Phong reflection model, which parameter controls how bright the specular highlight appears on a surface?

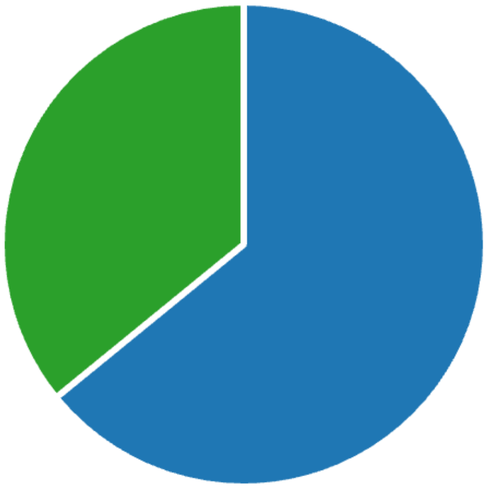
	Ambient reflection coefficient k_a	3
	Diffuse coefficient k_d	3
✓	Specular reflection coefficient k_s	22
	Specular exponent n, which raises (R · V) to a power	11



Vertex Shader

In a WebGL2 application, where is the most efficient place to apply model, view, and projection transformations to every vertex of all geometries in a 3D scene?

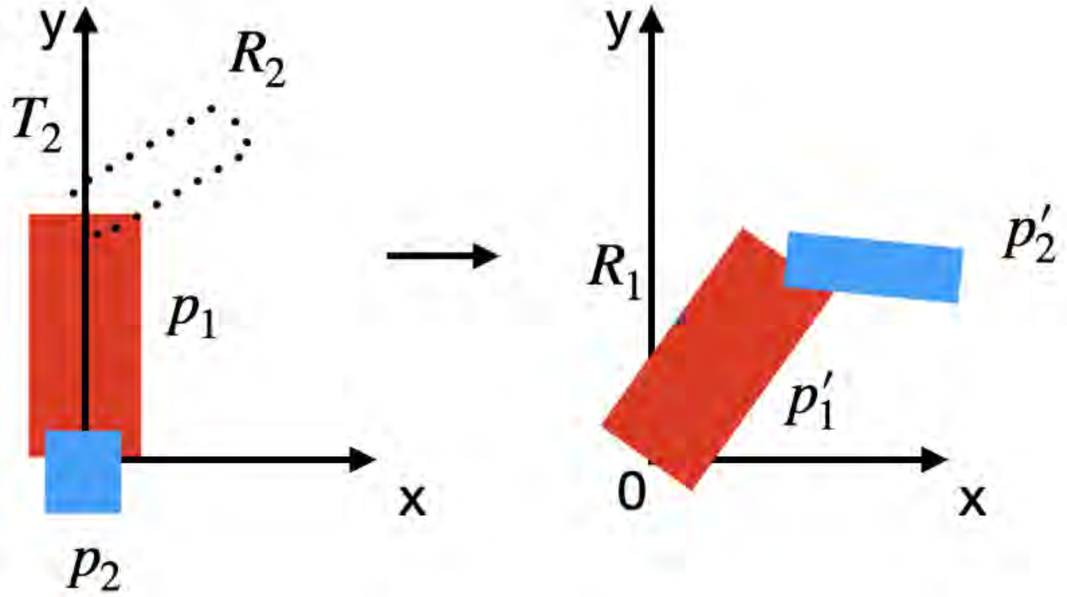
✓	Vertex shader	25
	Fragment shader	0
	In JavaScript on the CPU, by tra...	14
	for/while loop	0



Hierarchical Modeling

In the first step, the red cube on the right was obtained by applying the transformation $M_1 = R_1 \cdot T_1 \cdot S_1$ to a unit cube centered at the origin. Using hierarchical modeling, you will now transform the blue cube from P_2 (left) to P'_2 (right) with a transformation M_2. The dotted shape shows where the blue cube would be before the red cube's rotation R_1 is applied, and P'_2 is the result of applying R_1 to that dotted shape. What is M_2 expressed in the world frame?Notes:

1. Before M_1 was applied, the red cube was exactly the same size as the blue cube (both are unit cubes centered at the origin).
2. T_1 and T_2 are both translations along the y-axis of the form (0, t_i, 0), where t_i is half the height of the corresponding rectangle.



	$M_2 = R_1 \cdot T_1 \cdot S_1 \cdot R_2 \cdot T_2 \cdot S_2$	11
✓	$M_2 = R_1 \cdot 2T_1 \cdot R_2 \cdot T_2 \cdot S_2$	12
	$M_2 = M_1 \cdot R_2 \cdot T_2 \cdot S_2$	13
	$M_2 = R_1 \cdot R_2 \cdot T_2 \cdot S_2$	3



Lighting & Shading

A surface has normal **n** = (0, 1, 0), and the normalized direction to the light is **l** = (0, 0.6, 0.8). The light color is (1.0, 1.0, 0.5) and the material's diffuse color is k_d = (0.5, 0.8, 1.0). Only considering the Lambertian diffuse (no ambient or specular), what is the resulting RGB color at this point?

(0.30, 0.48, 0.30)	12
(0.50, 0.80, 0.50)	12
(0.30, 0.48, 0.60)	6
(0.40, 0.64, 0.40)	3
(0, 0.48, 0.40)	6



Lighting & Shading

What's your understanding of the diffuse term $K_d(n \cdot l)$ in Phong model? What are K_d , n and l in the diffuse term and how do they influence diffuse reflection?

The diffuse term models light that scatters equally in all directions from a matte surface

K_d is the diffuse reflection coefficient, describing how much of the incoming light the material reflects diffusely.

$\mathbf{n}$ is the unit surface normal, and $\mathbf{l}$ is the unit vector from the surface point toward the light.

Lighting & Shading

What is your understanding of parametric representation? And why we need to use it? Explain using a text description or a math formula (Latex code, etc.), or whichever is easier for you.

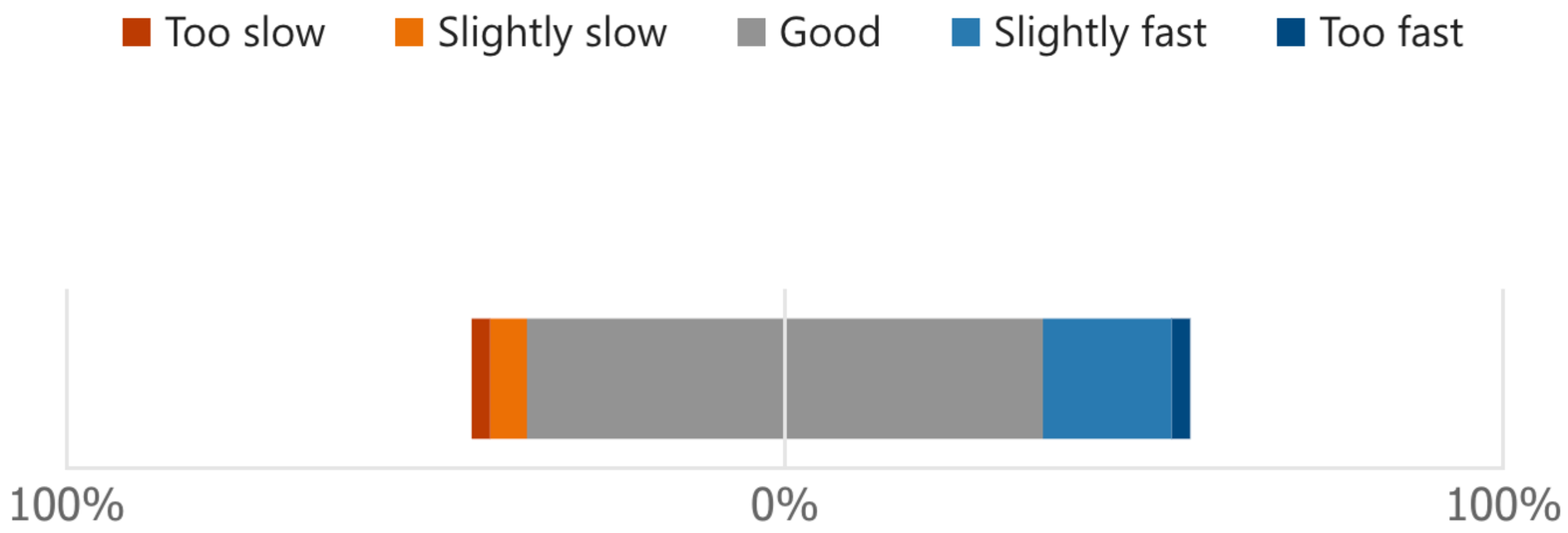
A parametric representation describes a curve or surface by expressing each coordinate as a function of one or more independent parameters.

A curve uses one parameter, $\mathbf{p}(t) = (x(t), y(t), z(t))$, and a surface uses two, $\mathbf{p}(u, v)$. For example, a circle of radius r can be written as $\mathbf{p}(t) = (r \cos t, r \sin t)$ for t in $[0, 2\pi]$. As the parameter sweeps through its range, it traces out every point on the shape.

Compared to the implicit form $f(x, y) = 0$, which is good for testing whether a point lies on or inside a shape, the parametric form makes it easy to generate points directly.

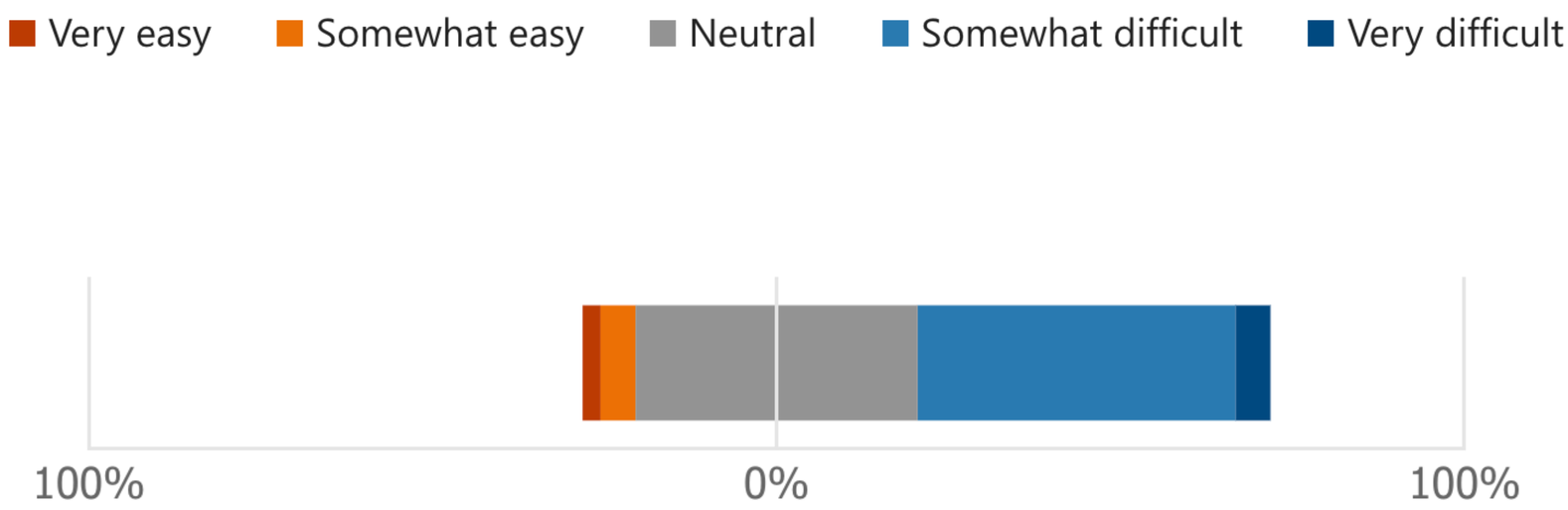
Course Feedback

How would you rate the pace of this class so far?



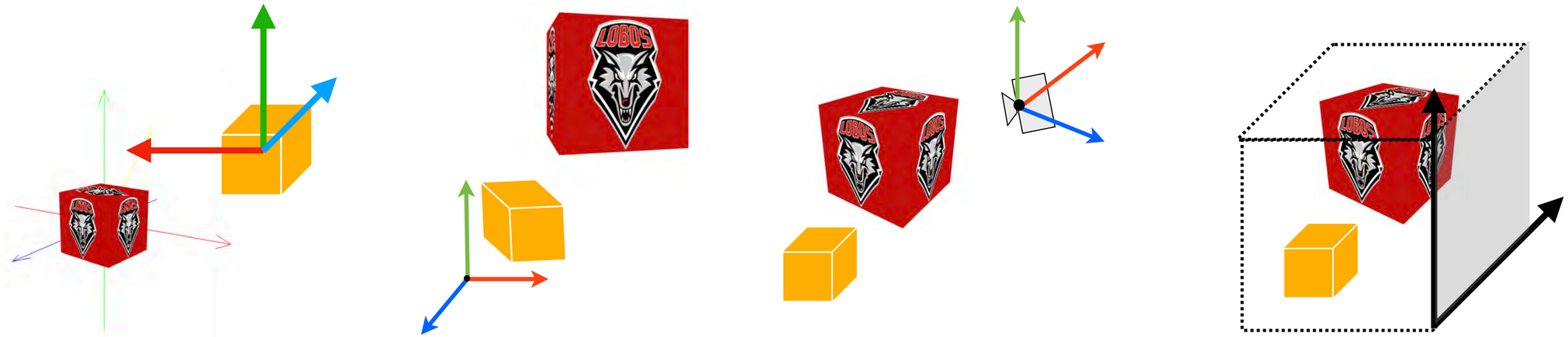
Course Feedback

How would you rate the overall difficulty of the course so far?



Shadings and Normals

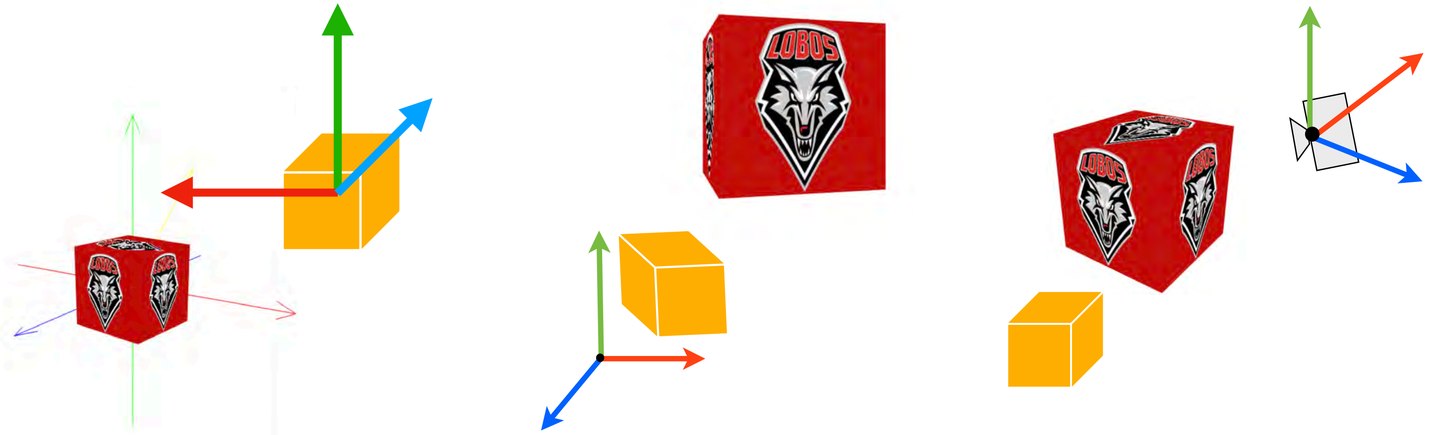
Shading Transformations



Most common to do it here

You can do it in any stage, but each has trade-offs.

Shading Transformations



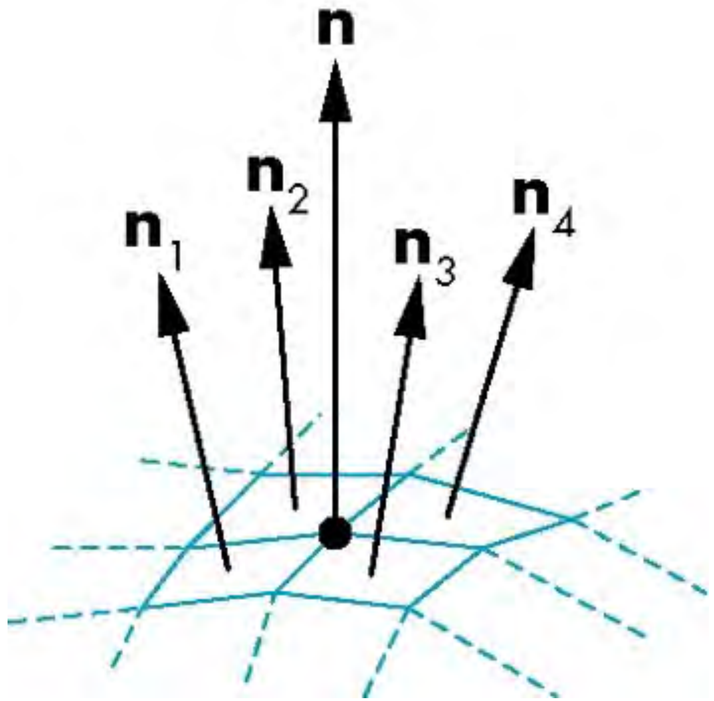
Model coordinates → Camera coordinates

$p' = Mp$ where M is model-view transformation

$n' = M_s n$ where M_s is shading transformation, it is usually 3x3 matrix

Averaging Normals for Smooth Shading

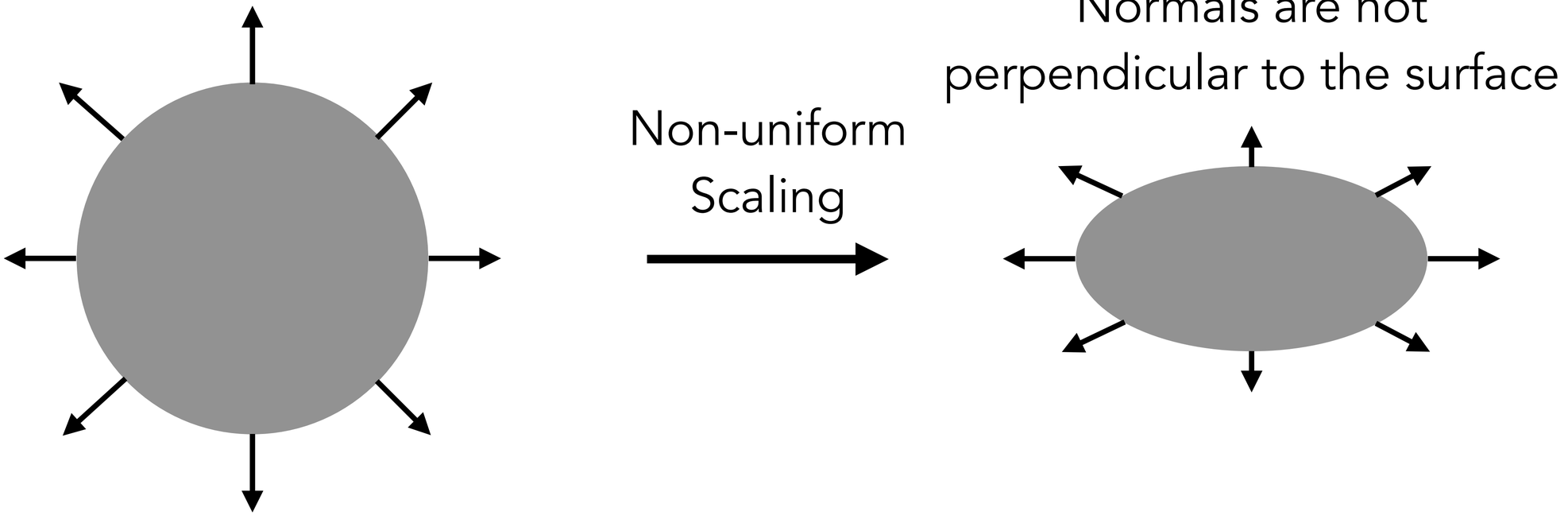
Smooth shading from an existing 3D model with flat shading



[source: Textbook by Prof. Ed Angel]

$$\mathbf{n} = \frac{\sum_i \mathbf{n}_i}{\left\| \sum_i \mathbf{n}_i \right\|}$$

When Scaling Breaks Your Normals



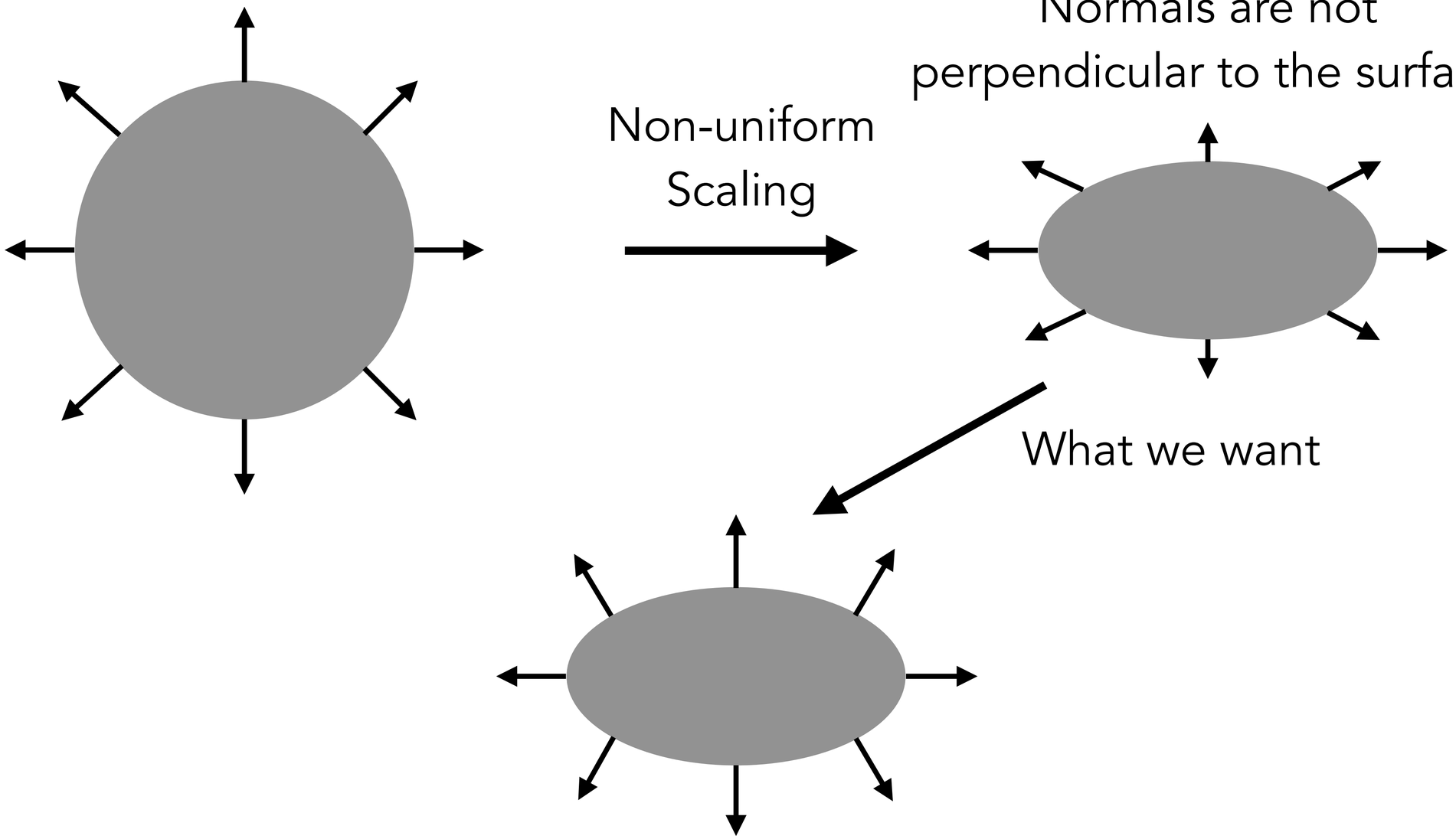
Non-Uniform Scaling Distorts Normals

Non-uniform scaling could cause the transformed normals no longer perpendicular to the surface.

Positions move with the surface. But normals must stay perpendicular to that surface after transformation.

So we need to apply some correction to normals.

Transforming Normals



Inverse Transpose Transformation

$$a \cdot (Mb) = a^T(Mb)$$

$n \cdot t = 0$ where t is a tangent vector lying on the surface

We want n' to satisfy

$t' = Mt$ ⟶ $n' \cdot t' = 0$

$$n' \cdot (Mt) = 0$$

$$(M^T n') \cdot t = 0$$

$$(M^T n') = n$$

$$n' = (M^{-1})^T n \qquad M_s = (M^{-1})^T$$

fundamental property of the dot product

$$a \cdot (Mb) = a^T(Mb) \qquad a \cdot (Mb) = a^T(Mb) = (a^T M)b = (M^T a)^T b = (M^T a) \cdot b$$

What About Arbitrary Transformations?

Singular Value Decomposition (SVD) $M = U\Sigma V^T$

$n \cdot t = 0$ where t is a tangent vector lying on the surface

When M, M_s are 3x3

$$t' = Mt = U\Sigma V^T t$$

We want n' and M_s to satisfy

$$n' \cdot t' = 0 \qquad M_s = R_1 S^{-1} R_2$$

$$M^{-1} = (R_1 S R_2)^{-1} = R_2^{-1} S^{-1} R_1^{-1}$$

$$\begin{aligned}(M^{-1})^T &= (R_2^{-1} S^{-1} R_1^{-1})^T \\ &= (R_2^T S^{-1} R_1^T)^T \\ &= R_1 (R_2^T S^{-1})^T \\ &= R_1 (S^{-1})^T R_2 \\ &= R_1 S^{-1} R_2\end{aligned}$$

Singular Value Decomposition (SVD)

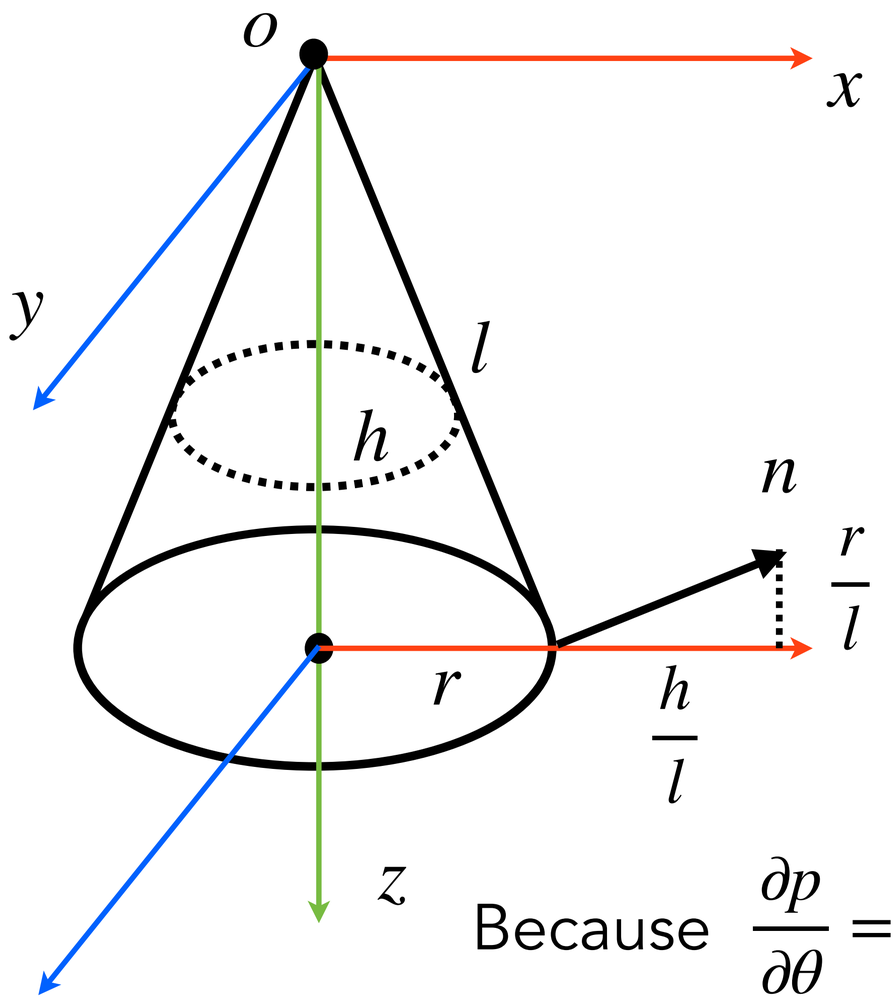
$$M = U\Sigma V^T$$

M can be decomposed as

$$M = R_1 S R_2$$

Geometries with Curvatures

Cone



Apex at the origin, base is on the plane $z = h$

$\theta \in [0,2\pi)$ the angle around z-axis

$u \in [0,1]$ the position moving along z-axis

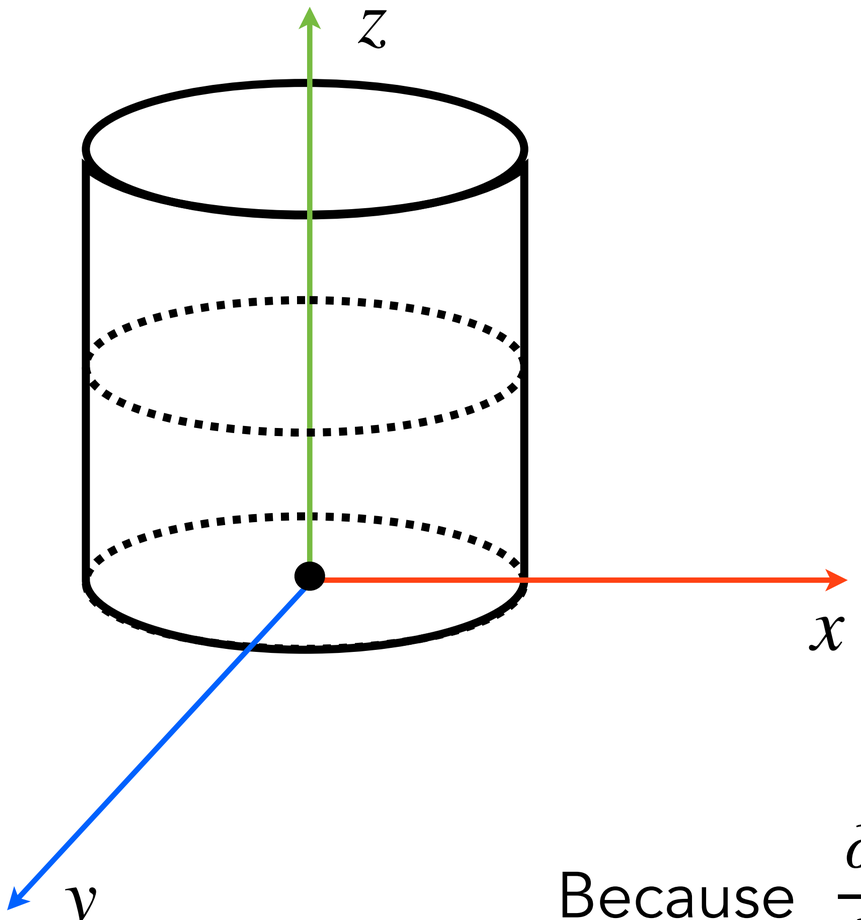
$$p(\theta, u) = \begin{bmatrix} rucos\theta \\ rusin\theta \\ h(1-u) \end{bmatrix}$$

normalized

$$\hat{n} = \frac{1}{\sqrt{r^2 + h^2}} \begin{bmatrix} hcos\theta \\ hsin\theta \\ r \end{bmatrix}$$

Because $\frac{\partial p}{\partial \theta} = \begin{bmatrix} -rusin\theta \\ rucos\theta \\ 0 \end{bmatrix}$ $\frac{\partial p}{\partial u} = \begin{bmatrix} rcos\theta \\ rsin\theta \\ -h \end{bmatrix}$ $n = \frac{\partial p}{\partial \theta} \times \frac{\partial p}{\partial u} = ru \begin{bmatrix} hcos\theta \\ hsin\theta \\ r \end{bmatrix}$

Cylinder



Apex at the origin, base is on the plane $z = h$

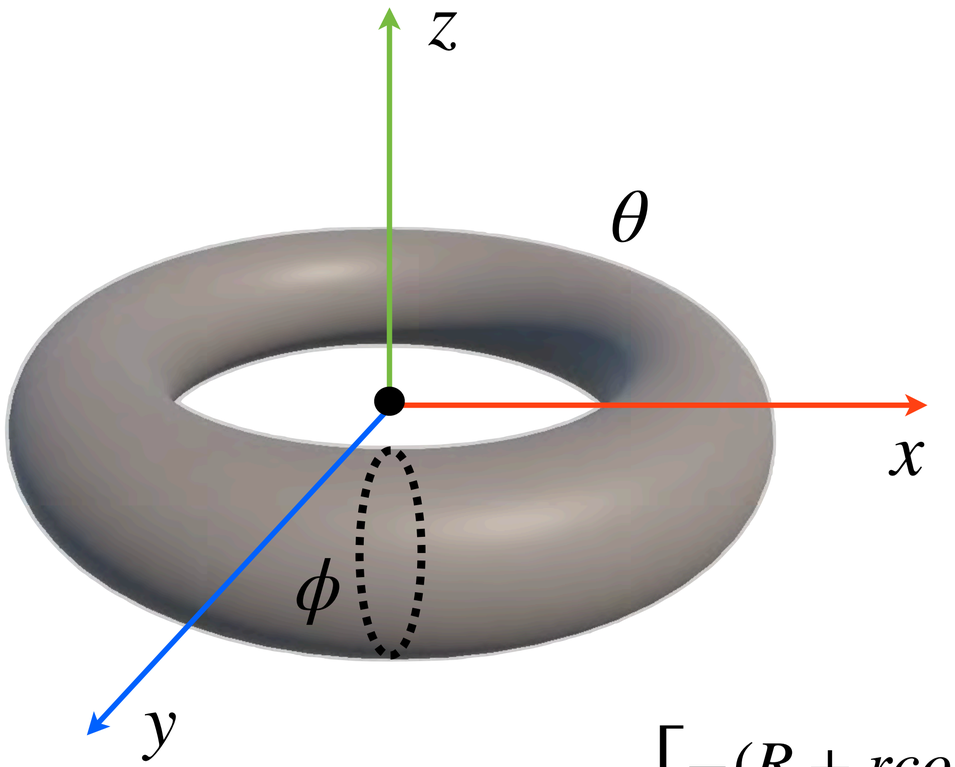
$\theta \in [0, 2\pi)$ the angle around z-axis

$z \in [0, h]$ the position moving along z-axis

$$p(\theta, z) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \\ z \end{bmatrix} \quad \text{normalized} \quad \hat{n} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix}$$

Because $\frac{\partial p}{\partial \theta} = \begin{bmatrix} -r\sin\theta \\ r\cos\theta \\ 0 \end{bmatrix}$ $\frac{\partial p}{\partial z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $n = \frac{\partial p}{\partial \theta} \times \frac{\partial p}{\partial z} = \begin{bmatrix} r\cos\theta \\ r\sin\theta \\ 0 \end{bmatrix}$

Torus/Donut



R large circle radius r tube radius

$\theta \in [0,2\pi)$ the angle around large circle

$\phi \in [0,2\pi)$ The angle around the tube

$$p(\theta,\phi) = \begin{bmatrix} (R + r\cos\phi)\cos\theta \\ (R + r\cos\phi)\sin\theta \\ r\sin\phi \end{bmatrix}$$

normalized

$$\hat{n} = \begin{bmatrix} \cos\phi\cos\theta \\ \cos\phi\sin\theta \\ \sin\phi \end{bmatrix}$$

Because $\dfrac{\partial p}{\partial \theta} = \begin{bmatrix} -(R + r\cos\phi)\sin\theta \\ (R + r\cos\phi)\cos\theta \\ 0 \end{bmatrix}$ $\dfrac{\partial p}{\partial \phi} = \begin{bmatrix} -r\sin\phi\cos\theta \\ -r\sin\phi\sin\theta \\ r\cos\phi \end{bmatrix}$ $n = \dfrac{\partial p}{\partial \theta} \times \dfrac{\partial p}{\partial \phi}$